Cologne Chip GateMate FPGA – CCGM1A1(20,480 CPEs), CCGM1A2(40,960 CPEs) or CCGM1A4(81,920 CPEs)

Free Bank Selection VDD 1.2V/1.8V/2.5V/3.3V User Pads Supply

IO North

IO East

IO South

IO West

{Ext-BUS}

{Main-BUS}

{Peri-BUS}

CCGM1A1(FBGA-324)

Table 3.1: Configuration mode setup

CFG_MD[3:0]	Configuration mode
0x0	0b 0000 SPI Active Mode CPOL = 0, CPHA = 0
0x1	0b 0001 SPI Active Mode CPOL = 0, CPHA = 1
0x2	0b 0010 SPI Active Mode CPOL = 1, CPHA = 0
0x3	0b 0011 SPI Active Mode CPOL = 1, CPHA = 1
0x4	0b 0100 SPI Passive Mode CPOL = 0, CPHA = 0
0x5	0b 0101 SPI Passive Mode CPOL = 0, CPHA = 1
0x6	0b 0110 SPI Passive Mode CPOL = 1, CPHA = 0
0x7	0b 0111 SPI Passive Mode CPOL = 1, CPHA = 1
0x8	0b 1000 JTAG Mode

* Modes not mentioned in the list may not be selected and lead to a malfunction of the device.

Table 3.2: Configuration process state

CFG_DONE	CFG_FAILED	Description
0	0	Invalid state
0	1	Configuration procedure is still in progress Unable to load configuration. Data streams might be invalid
1	0	Configuration successful
1	1	Invalid state

* Only tag CFG_FAILED is active. Short pulses have no significance.
** CFG_DONE might be set to 1 if the configuration procedure has to wait for user mode.

CFG_SET1 OFF ON

CONFIGURATION

JTAG

SPI

SERDES

Unequal Pins A1/A2/A4!
Ball V15 is SER_RTERM1 in A2/A4!

Power-Good Reset.

Holds #RST for 350ms(typical)

Tdelay(2.7V)=370ms
Voltage Supervisor with Open-Drain!
NJM2823 / HX7027(SOT-23-3) or VDA7210NTA

GPIO20\JTAG_RST

POR_EN

POR_RST1

POR_EN1

MCP120T-300I/TT(SOT-23-3)
Open-Drain, No Internal pull-up!

POR_EN – Enables power-on reset POR_EN has an internal pull-down resistor.
O: Reset state depends only on the POR_EN signal.
1: Reset state depends on the POR_EN signal and also when RST_N is 1 and POR indicates VDD and VDD_CLK voltages are stable.

SER_RX_P V12
SER_RX_N V13
SER_TX_P J13
SER_TX_N J12

Note6 Take in mind that Bank5 is not working when VDD_CORE = 0.9V (Low power mode), because VDDSER and VDDSER_PLL have got minimum working voltage: 0.95V!

CCGM1A1 Characteristics
VDDIO = 1.1-2.7V (I/O supply voltage (GPIO banks and VDD_CLK))
VDDcore = 0.85-1.15V (Core supply voltage (VDD, VDD_PLL, VDD_SER and VDD_SER_PLL))
VDDSER = 0.95-1.15V (SerDes supply voltage (VDD_SER))
VDDSER_PLL = 0.95-1.15V (SerDes PLL supply voltage (VDD_SER_PLL))

Cologne Chip

Power

Unequal Pins A1/A2/A4!
Ball U16 is SER_TX_P1 in A2/A4!
Ball V16 is SER_TX_N1 in A2/A4!
Ball U14 is SER_RX_P1 in A2/A4!
Ball V14 is SER_RX_N1 in A2/A4!
Ball N16 is VDD_SER in A4!

FPGA Oscillator

{Main-BUS}

CLKO_EN1

CLKO

FPGA Button&LED

{Main-BUS}

FPGA_BUT

FPGA_LED

USB PHY

{Peri-BUS}

USB-C

USB_2_D-

USB_2_D+

USB Diff. Pair Zdiff=90 Ohms!

TUSB1106RWR

USB-C-16P-F(A40-00119-A52-12)

Power Supply

Power-On Sequence:
VDD_CORE -> +2V5 -> +3V3 -> +1V8 -> +1V2

Note6 Take in mind that SerDes is not working when VDD_CORE = 0.9V (Low power mode), because VDDSER and VDDSER_PLL have got minimum working voltage: 0.95V!

Vout=0.6*(1+Ra/Rb)

TPS62402ADRLR(SOT-563-6)

H-WK-1x3-LF(1-2=Closed/2-3=Opened)

VDD_CORE_SET1	VDD_CORE (Mode)	fDCO (MHz)	fPLL (MHz)
Both Opened	0.90V (Low power mode)	500-1000	up to 250
1–2 Closed	1.00V (Economy mode)	1000-2000	up to 312.5
2–3 Closed	1.10V (Speed mode)	1250-2500	up to 416.75

PG_VCore

PG_2V5

PG_3V3

PG_1V8

PG_1V2

2*32M Bits Serial Pseudo-SRAM with QPI

Clock rate: 100MHz(max) for fast read

{Peri-BUS}

LY68L3200SLT(SOIC-8_150mil)

Mounting Holes

MH2 MH3 MH4 MH5

JTag Flasher

{USB-BUS}

USB-C-16P-F(A40-00119-A52-12)

USB_D-

USB_D+

USB Diff. Pair Zdiff=90 Ohms!

QSPI pins of RP2040 should be wired directly to the flash, using short connections

Flash Memory – 16Mbit(2MB)
VCC = 1.65V to 3.6V!

W25Q16JVSSIQ(SOIC-8_208mils)

RP2040 Characteristics
IOVDD = 1.8-3.3V
DVDD = 1.1V
VREG_VIN = 1.8-3.3V
USB_VDD = 3.3V
ADC_AVDD = 3.3V

BOOTSEL

RP2040_RST#
FPGA_RESET_IN

RP2040_SWCLK
RP2040_SWDIO

Close to RP2

RP2040(QFN-56_7x7mm_P0.4mm)

{Main-BUS}

{USB-BUS}

RP2040_SWCLK

RP2040_SWDIO

PGM/DBG

GPIO17_DBG-UART_TX

GPIO18_DBG-UART_RX

Extensions

{Ext-BUS}

{Main-BUS}

UEXT & PMOD Connectors

{Ext-BUS}

TXB0108PWR(TSSOP-20)

Bidirectional Voltage-Level Translator. Data Rate(Max) = 100Mbps

Vout	EA_B7	EA_A7
5V	LOW	–
3V3	HIGH	Z
1V8	HIGH	LOW

ISO/IEC 7816 Interface

{Ext-BUS}

Notes:
– Supports 3 classes: A,B,C
– Different Supply Voltages: 5V, 3.3V, 1.8V
– three AS – Application Specific pins
– does not support +21V DC [Programming] if used as Master:
– Place all components
– 5V direct pass though of the USB Voltage
– 3.3V, 1.8V 500mA current limited

TXB0108PWR(TSSOP-20)

https://www.olimex.com/
based on GateMateA1-EVB

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